a

Semantic Isolation by Disease Category

Semantic Isolation Index (SII)

0.0030
0.0025
0.0020
0.0015
0.0010

NTDs
Injuries
Mental disorders
Infectious
Neurological
Cardiovascular
Musculoskeletal
Other
Neoplasms
Respiratory
Digestive

b

Top 20 Most Semantically Isolated Diseases (injury causes excluded)

Conduct disorder
African trypanosomiasis
Onchocerciasis
Ebola
Zika virus
Lymphatic filariasis
Guinea worm disease
Fungal skin diseases
Chagas disease
Cystic echinococcosis
Attention-deficit hyperacti...
Yellow fever
Leishmaniasis
Inguinal, femoral, and abdo...
Anxiety disorders
Invasive Non-typhoidal Salm...
Trachoma
Cysticercosis
Intestinal nematode infections
Iodine deficiency

WHO NTD
Global South (non-NTD)
Other

9/20 WHO NTDs
15/20 Global South

0.000
0.001
0.002
0.003
0.004
0.005

Semantic Isolation Index (SII)

c

Semantic Isolation vs Research Volume

r = -0.64
P < 0.001

Other
Global South
NTD

African trypanoso...
Onchocerciasis
Lymphatic filariasis
Guinea worm disease
Chagas disease

Semantic Isolation Index (SII)

0.0030
0.0025
0.0020
0.0015
0.0010

2.5
3.0
3.5
4.0
4.5
5.0
5.5

Publication Count ($\log_{10}$)

d

NTDs Show Significantly Higher Isolation

***

+45% higher
P < 0.0001
d = 1.80

Semantic Isolation Index (SII)

0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010

NTDs
(n=15)

Other diseases
(n=160)